Plain S_5 C_4 with the D_4 oracle's solver options

Question. `src/cryptanalysis/jacobian_transfer/cost.rs` charged every `k = 4` extension index-calculus decomposition $C_4 = 1.227 \cdot 10^{12}$ `F_p` multiplications: §11.17's plain S_5 solve (`research/notes/index-calculus/RESEARCH_RESIDUAL_WALKS.md`), measured before #1792 added sparsest pivots, the chain eigen step and shape replay to the same Macaulay solver for the D_4-symmetric oracle (where they cut C_4 about 370×, 26× of it from sparsest pivots). Do those options, which do not depend on the D_4 form, also cut the plain S_5 solve?

Answer: yes, 13.9×. On §11.17's own instances the plain solve costs $C_4 = 8.69 \cdot 10^{10} \approx 2^{36.34}$ with all three options (mean over 12 replayed decompositions, the one replay fallback charged), against $1.209 \cdot 10^{12}$ for the committed solver on the same residuals. `cost.rs` now charges $8.69 \cdot 10^{10}$. No route class and no curve verdict in `docs/curves/jacobian-transfers.json` changes: the registry has no `k = 4` extension index-calculus row.

Class (AGENTS.md "Success criterion"): an **engineering** improvement of a stage (C_4, one decomposition), measured on counted `F_p` multiplications. It is not an end-to-end `S`; no `k = 4` plain pipeline exists to move.

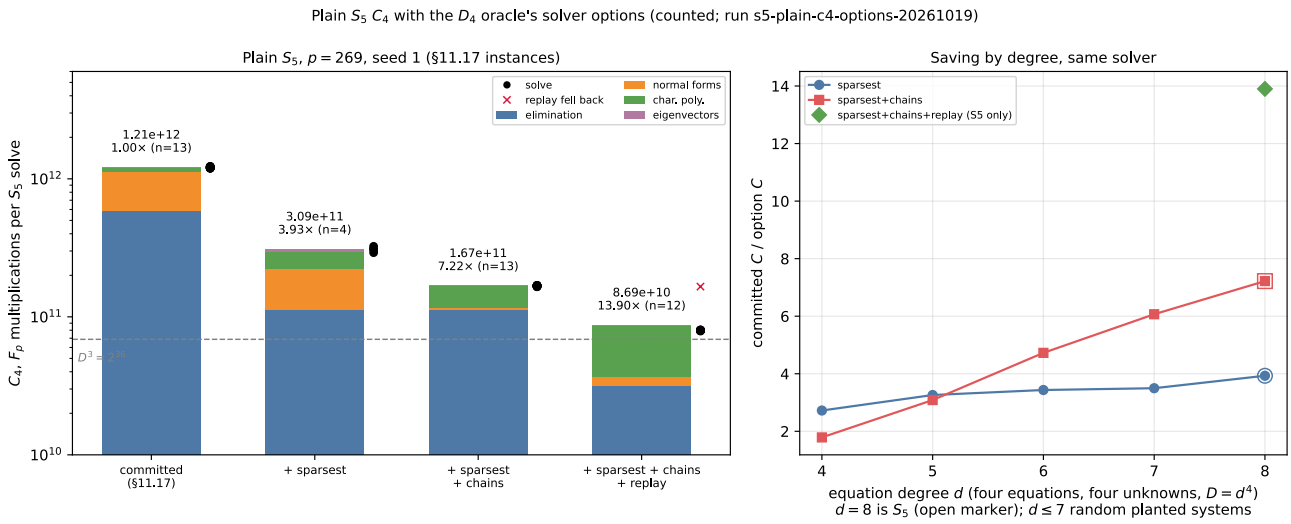


Figure 1: C4 by option

Setup

- **Solver.** `gaudry_quartic::solve_system_w_inner` with weights `(1,1,1,1)` (the plain system has no $\sigma_4 = \pi^2$ to substitute, so #1792's `weighted` option does not apply). The new public wrapper `solve_s5_subspace_with` / `solve_system4_with` (`PlainOptions { chains, sparsest }` plus an optional `ShapeTrace`) is, with options off, the committed `solve_s5_subspace` count for count (unit test `plain_options_keep_the_solutions_and_off_is_the_committed_solve`, and the committed arm below reproduces §11.17's counts exactly).
- **Arms.** `committed` (§11.17); `sparsest` (pivot = sparsest candidate row); `sparsest+chains` (eigen step along the chains of e_1); `sparsest+chains+replay` (F4Remake: the first solve records which rows reduce to zero, later solves skip them and top up from the held rows if the rank falls short; a shape mismatch runs the full solve, both charged).
- **Instances.** `p = 269`, seed 1: §11.17's curve, 149-point factor base and residual stream. Residual 0 is the constructed four-point sum; the residual stream advances with the committed arm's solve, as it did in §11.17, so residuals 0–3 are §11.17's four.
- **Pairing and order.** Every arm of a residual starts from a clone of the same generator state. The arm order rotates by one per residual (a Latin square over the 4-arm run; each arm first, second and third over the 3-arm run).
- **Correctness.** Every solve's decomposition set is compared with the meet-in-the-middle oracle.
- **Unit.** `F_p` multiplications, counted as §11.17 counts them (elimination, normal forms, characteristic polynomial, roots, eigenvectors, Weil restriction, splitting). Counts do not depend on contention.
- **Host.** Apple M4 Pro, 14 cores, 48 GiB, macOS, `RAYON_NUM_THREADS=4`, no isolation; load average about 26 from other sessions' jobs (`raw/host.txt`). Wall times are recorded but contended, so they are descriptive only.

Results

S_5, residuals 0–12 (`raw/s5_p269_seed1_r13.json`)

arm	n	mean C_4	$\log_2$	committed / arm, ratio of means	paired ratio, min–max	oracle-equal
committed	13	$1.2093 \cdot 10^{12}$	40.14	1	—	13/13
sparsest + chains	13	$1.6750 \cdot 10^{11}$	37.29	**7.22**	7.12–7.39	13/13
sparsest + chains + replay	12	**$8.690 \cdot 10^{10}$**	36.34	**13.90**	7.24–15.37	13/13

The replay arm's first solve (residual 0, $1.689 \cdot 10^{11}$) records the shape and is excluded from its mean, as #1792 excluded it. Of its 12 later solves, 11 replayed (7.89–$8.04 \cdot 10^{10}$) and one, residual 3, fell back to the full solve ($1.656 \cdot 10^{11}$; the triangularised leading monomials differed, the only check before elimination, so the failed attempt cost almost nothing). The mean charges that fallback.

Phase means (replay arm): elimination $3.17 \cdot 10^{10}$, normal forms $5.20 \cdot 10^{9}$, characteristic polynomial $4.97 \cdot 10^{10}$, eigenvectors $2.9 \cdot 10^{8}$. Against the committed solver, elimination and normal forms ($5.82 \cdot 10^{11}$ and $5.46 \cdot 10^{11}$) no longer dominate: the characteristic polynomial of the chain step, $0.72 \cdot D^3$ with $D = 4{,}096$, is now 57 % of C_4.

Ablation, residuals 0–3 (`raw/s5_p269_seed1.json`, 4 arms, Latin square)

arm	n	mean C_4	committed / arm	paired min–max
committed	4	$1.2130 \cdot 10^{12}$	1	—
sparsest	4	$3.0865 \cdot 10^{11}$	**3.93**	3.78–4.11
sparsest + chains	4	$1.6758 \cdot 10^{11}$	7.24	7.21–7.26
sparsest + chains + replay	3	$1.0855 \cdot 10^{11}$	11.13	7.24–15.16

Same instances and counts as the first four residuals above (each count reproduced exactly between the two runs). Sparsest pivots alone give 3.9× here, not the 26× they gave on the D_4 system's plain five-unknown form: that elimination's fill-in was far worse.

Replay checked on nonempty decompositions (`raw/s5_p269_seed1_constructed4_replay.json`)
In the paired runs only residuals 0 and 6 decompose. To check replayed solves on decompositions that exist, four constructed residuals (planted four-point sums) were solved by the replay arm alone: the recording solve and all three replays (7.94–$7.97 \cdot 10^{10}$) return exactly the planted quadruple, equal to the oracle.

Generic systems, same solver (`raw/generic_d4-7.json`)
Four random planted equations of degree d in four unknowns, as §11.17′s scaling panel (committed counts reproduce §11.17′s: $2.30 \cdot 10^{8}$, $3.66 \cdot 10^{9}$, $3.55 \cdot 10^{10}$, $2.40 \cdot 10^{11}$). Committed / option:

d	sparsest	sparsest + chains
4	2.72	1.79
5	3.26	3.09
6	3.44	4.73
7	3.50	6.07
8 (S_5, above)	3.93	7.22

The chain step's saving grows with d; S_5 sits on the generic trend, as §11.17 found for the committed solver.

What changed in the cost model

- `extension_measured_solve_mults`: `(4, 1.227e12)` ⟶ `(4, 8.69e10)`, i.e. $\log_2 C_4$ 40.16 ⟶ 36.34 (–3.82 bits on every `k = 4` extension IC price).
- `log2_extension_solve_mults` charged `max(measured, D³)`. D^3 is not a floor: the chain step alone costs $0.72 \cdot D^3$ here, and #1792's D_4 oracle measured about `2^26` against the plain D^3 `= 2^36` (and below its own quotient's 512^3 `= 2^27`). A measured `C_k` is now charged as measured; D^3 stays the estimate where nothing is measured (`k ≠ 3, 4`), described as an estimate, not a lower bound, and labelled `degree_cubed_estimate` (was `degree_cubed_floor`) in the registry. The test asserts that a measured price below D^3 is not raised. With $8.69 \cdot 10^{10}$ `= 2^36.34` the clamp would not have bound today; it would have at any further 1.27×.
- `docs/curves/jacobian-transfers.json` regenerated once (`jacobian_transfers build --budget 200000`, `check` current). Verdicts before and after: 598 `formula_advantage`, 57 `formula_marginal`, 1,917 `boundary` routes; every route class and every

curve verdict identical (diffed route by route). Only the notes, the basis label and the cost-model block changed. **No boundary verdict changed**, because the registry's extension index-calculus rows are at k = 2, 3, 5, 6, 10, 11 and 14, never 4.

Derived, not measured

On §11.16′s formula $p^* = 0.095 \cdot C_4 / (1 - r_\infty)$ and its exponents, a factor 13.9 off C_4 moves the crossover n^* about $4 \log_2 13.9 \approx 15$ bits earlier: from 2^149.4 to about 2^134 at $r_\infty = 0.34$. This is an extrapolation from a stage cost, as §11.17′s was.

Limits

- One prime (p = 269), one curve; 13 paired residuals, 11 replayed solves of the arm that sets the price. Counts vary 2.3 % across residuals with the committed solver (eigenvector step) and under 2 % within the replayed set.
- The replay fallback rate (1 of 12) is a small sample; at a fallback rate f the mean is about $(1 - f) \cdot 7.96 \cdot 10^{10} + f \cdot 1.66 \cdot 10^{11}$.
- Not compared with an external solver (msolve, Magma F4 + sparse FGLM) on the same system (AGENTS.md §15). The price is this repository's best verified solver, not the fastest known.
- Wall time contended (host load about 26): descriptive only. Counted multiplications are the result.

Reproduce

```
cargo run --release --example gaudry_quartic_c4_options -- --p 269 --seed 1 --residuals 13 \
    --arms committed,sparsest+chains,sparsest+chains+replay --json raw/s5_p269_seed1_r13.json
cargo run --release --example gaudry_quartic_c4_options -- --p 269 --seed 1 --residuals 4 \
    --arms committed,sparsest,sparsest+chains,sparsest+chains+replay --json raw/s5_p269_seed1.json
cargo run --release --example gaudry_quartic_c4_options -- --p 269 --seed 1 --residuals 4 --constructed 4 \
    --arms sparsest+chains+replay --json raw/s5_p269_seed1_constructed4_replay.json
cargo run --release --example gaudry_quartic_c4_options -- --generic 4,5,6,7 \
    --arms committed,sparsest,sparsest+chains --json raw/generic_d4-7.json
cargo run --release --example gaudry_quartic_c4_options_summary -- raw/s5_p269_seed1_r13.json - raw/
summary_r13.json
cargo run --release --example gaudry_quartic_c4_options_summary -- raw/s5_p269_seed1.json raw/
generic_d4-7.json raw/summary.json
MPLCONFIGDIR=$TMPDIR python3 figures/plot.py
```

raw/generic_d4-6.json is the first pilot of the generic panel (identical counts to the d ≤ 6 rows of generic_d4-7.json). Run report: reports/run-20261019T1851Z/. Report PDF: REPORT.pdf.